Data Science Certificate Diploma
DS650 Data Analytics

Homework 3 (20 points)

You are expected to deliver a PDF document (this document) and one Jupyter notebook with your Python code.

For the following questions, we will be working with an SMS Spam dataset provided as a .csv file. The dataset contains SMS messages labeled as either "spam" or "ham" (not spam). Our goal is to analyze, preprocess, and build a machine learning model to classify messages as spam or not. All questions must be answered using Google Colab.

To obtain reproducible results, call the following command in the beginning of your code:

```
np.random.seed(123)
```

a. Read the data into a data frame. Report in the table below the total number of spam and ham messages in the dataset. What do you observe? Which evaluation metric might be more useful (e.g. accuracy)? (2 points)

Label	**Count**
Ham	4825
Spam	747

Answer: We see clearly we have imbalance between our ham and spam classes, ham been much higher than spam so accuracy alone is not a good measure of model performance we need to delve deeper using precision, recall and f1-score.

b.
We will convert the messages into tokens with different ways. Firstly, use the `LemmaTokenizer` class (that tokenizes the text, lemmatizes each word, and returns the cleaned tokens), the `StemTokenizer` class and the `get_wordnet_pos` function(maps POS

tags to WordNet tags for lemmatization) from the count `vectorizer.ipynb` file from Activity Week 5 Folder. Secondly, test these two tokenizers on the sample message (displaying in the box below the tokens created for each method):

sample_msg = ["Congratulations you have won a 1000 dollars gift card Click here to claim your prize"]

(2 points)
Answer:

```
Lemma Tokens: ['Congratulations', 'you', 'have', 'win', 'a', '1000', 'dollar', 'gift', 'card', 'Click', 'here', 'to', 'claim', 'your', 'prize']
Stem Tokens: ['congratul', 'you', 'have', 'won', 'a', '1000', 'dollar', 'gift', 'card', 'click', 'here', 'to', 'claim', 'your', 'prize']
```

c. Let's return to on our data! Create a binary label column called b_labels where 'ham' is mapped to 0 and 'spam' is mapped to 1. Split the dataset into training and testing sets (80% train, 20% test). Then convert the text data in the train and test set into numerical vectors using 5 different techniques:

- CountVectorizer with default arguments
- CountVectorizer with lemmatization
- CountVectorizer with stemming
- TF-IDF vectorizer with default arguments
- TF-IDF vectorizer with a maximum of 2000 features

Report and compare the shape of the resulting matrices for the training set.

(# of rows, # of columns)

(3 points)

Vector Model	**Matrix Shape**	
	Train	**Test**
TFIDF	(4457, 7682)	(1115, 7682)
TFIDF with 2000 features	(4457, 2000)	(1115, 2000)
Count Vectorizer	(4457, 7682)	(1115, 7682)
Count Vectorizer with Lemmatization	(4457, 7500)	(1115, 7500)
Count Vectorizer with Stemming	(4457, 7204)	(1115, 7204)

Answer:
Default vectorizer and TFIDF both have a large number of features with lemmatization we get fewer and stemming reduces them bit more. This dimmensionality reduction translates to thte correct grouping of related words.

c. Train a Logistic Regression classifier using with all the above vectorizers as input features. Instead of simply training your models on the entire training set and evaluating on the test set, we will use 5-fold cross validation which is a methodology introduced in Lecture 2 for model evaluation and selection. For each vectorizer use 5-fold cross validation with grid search for selecting the hyperparameter C with options [1, 0.1, 0.01] and F1-score as the metric. Report the cross validation F1-score of these models.
(5 points)

Vector Model	**CV F1-score Logistic Regression**
TFIDF	0.8203
TFIDF with 2000 features	0.8641
Count Vectorizer	0.9305
Count Vectorizer with Lemmatization	0.9261
Count Vectorizer with Stemming	0.9284

d. Which preprocessing technique works best? What is the optimal value for C? (1 point)

Answer: Comparing the f1-scores of each technique we conclude that CountVectorizer with default configuration provides the best approach and best C value is 1. We deduce that lemmatization and stemming remove words that are include important information and that tf-idf weighting did not provide any advantage in this specific dataset and problem.

e. Choosing the TF-IDF with 2000 features and Count Vectorizer with Stemming from above use 5-fold cross validation with a random forest classifier using F1 score as a metric with the following hyperparameters for tunning:
'n_estimators': [50, 100, 200],
'max_depth': [None, 10, 20],
'min_samples_split': [2, 5, 10]

Report the cross validation F1 score and the best hyperparameters.

(2 points)

Vector Model	**CV F1-score Random Forest**
TFIDF with 2000 features	0.9267
Count Vectorizer with Stemming	0.8947

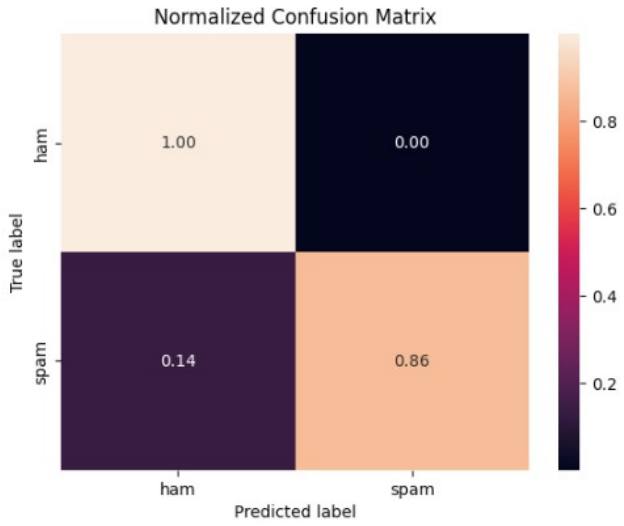
Answer: Tf-idf has a better f1-score suggesting that weighting scheme and with a limited vocabulary size was more effective using a Random Forest model. The best hyperparameter are min_samples_split of 10, no max depth on our trees and 200 estimators according the results of our code.

f. Select the best model among random forest and logistic regression and evaluate the model on the test set and report the accuracy, precision, recall, and F1-score. Plot in the following box the normalized confusion matrix based on the test data.

(4 points)

Metric	**Best Model Test Set**
Accuracy	0.9830
Precision	0.9914
Recall	0.8647
F1-score	0.9237

Normalized confusion matrix



g. Examine false positives (ham classified as spam) and false negatives (spam classified as ham) from the test set. Show those misclassified cases in the box below and elaborate on what might have confused the algorithm.

(3 points)

Answer: In the false positives and negatives report we see that contain :

- unusual characters or symbols
- some typos and misspellings
- words or phrases that were probably not present in our training data
- urls

All the above can interfere with the accuracy of our model

```
False Positives (Ham Classified as Spam):
3690    Sir, i am waiting for your call, once free ple...

False Negatives (Spam Classified as Ham):
671            SMS. ac sun0819 posts HELLO:\You seem cool
730     Email AlertFrom: Jeri StewartSize: 2KBSubject:...
1448    As a registered optin subscriber ur draw 4 å£1...
1268    Can U get 2 phone NOW? I wanna chat 2 set up m...
2769    Burger King - Wanna play footy at a top stadiu...
5370    dating:i have had two of these. Only started a...
2246    Hi ya babe x u 4goten bout me?' scammers getti...
1673    Monthly password for wap. mobsi.com is 391784....
2401    Babe: U want me dont u baby! Im nasty and have...
3130    LookAtMe!: Thanks for your purchase of a video...
2351    Download as many ringtones as u like no restri...
4014    You will be receiving this week's Triple Echo ...
606     XCLUSIVE@CLUBSAISAI 2MOROW 28/5 SOIREE SPECIAL...
5449    Latest News! Police station toilet stolen, cop...
1499    SMS. ac JSco: Energy is high, but u may not kn...
528     You will recieve your tone within the next 24h...
4071    Loans for any purpose even if you have Bad Cre...
2821    ROMCAPspam Everyone around should be respondin...
```